A3-PS028-TC278

Happy Eyeballs IPv4/IPv6 Fallback Measurement

Comparing curl and Chromium in a controlled dual-stack lab

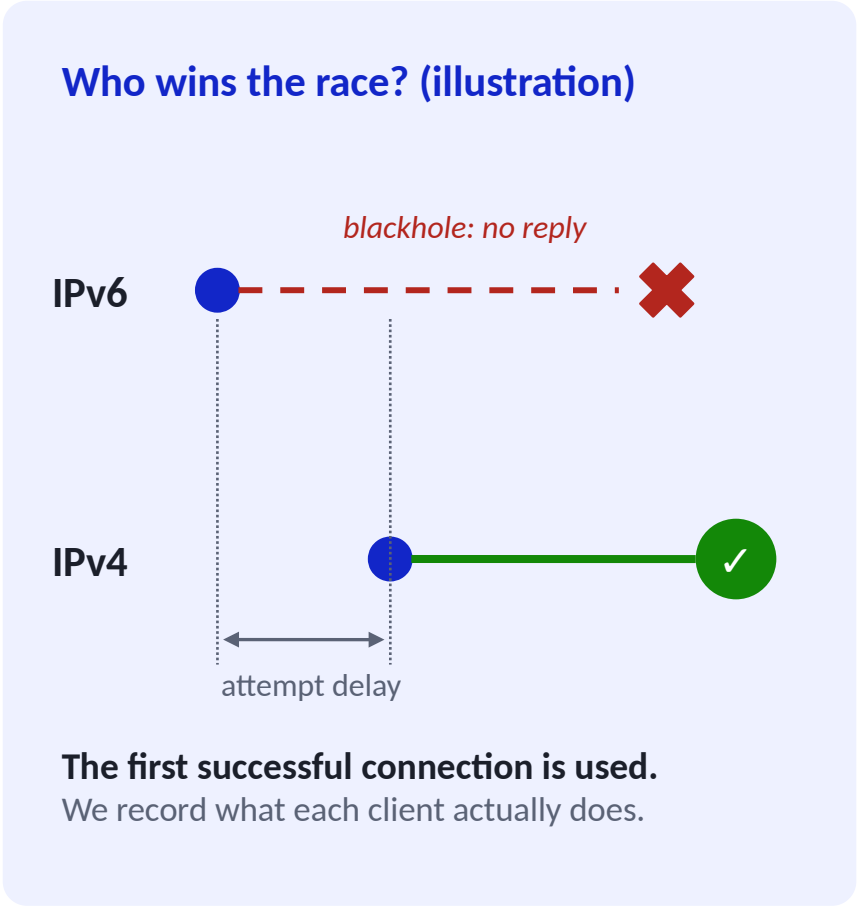
NetDefenders

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AIORI Hackathon 3

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Problem Description

A slow or broken IPv6 path can delay a connection; we measure how real clients recover

1. **Slow or broken IPv6 can delay a connection**
   Without timely fallback, a client that tries addresses one at a time may wait for TCP retransmission or connection timeouts before trying another address.

2. **Happy Eyeballs staggers the attempts**
   RFC 8305 [1] staggers attempts across address families and uses the first connection that succeeds.

3. **Implementations make their own choices**
   RFC 8305 recommends values, but clients and operating systems decide what to implement [6].

4. **Prior work shows why measurement matters**
   Existing studies and tools already measure Happy Eyeballs behaviour [5], [6], [7].

Our contribution: a compact, reproducible dual-stack testbed for controlled failure scenarios, with per-run packet captures and a transparent comparison of curl and Chromium.

2 existing clients: curl and Chromium

6 failure and delay scenarios, plus a healthy control

≥20 repetitions per condition per client

≥280 runs in total, each with a packet capture

Background

How a dual-stack client chooses between IPv6 and IPv4

Dual-stack hosts

DNS can return both A (IPv4) and AAAA (IPv6) records. RFC 8305 [1] assumes host policy favours IPv6; RFC 6724 [3] sets the address-selection rules, so the real order depends on local policy and reachability.

Happy Eyeballs

RFC 6555 [2] introduced racing connection attempts. RFC 8305 [1] (version 2) added asynchronous DNS, address interleaving and staggered attempts.

Version 3 draft

draft-ietf-happy-happyeyeballs-v3-04 [4] adds support for SVCB/HTTPS DNS records, which give alternative endpoints and protocol hints such as ALPN. Still an Internet-Draft: background only.

One RFC 8305 connection setup

1 Resolve
Send AAAA and A queries together. If A returns first, wait briefly for AAAA (50 ms recommended).

→

2 Order
Sort by address-selection policy (RFC 6724); interleave the two families.

→

3 First attempt
Start with the first address; policy and client decide which family that is.

→

4 Stagger
Start further attempts after the Connection Attempt Delay (250 ms is one recommended default).

→

5 Use success
Use the first successful connection; cancel pending attempts where appropriate.

Values are RFC 8305 recommendations; clients may differ, which is what we measure. RTT = round-trip time. SYN, SYN-ACK and ACK = the three packets of a TCP handshake.

Literature & RFC Review

Standards recommend values; measurement studies show implementations make their own choices

Source	Key contribution	How we use it
RFC 8305 [1] Happy Eyeballs v2, 2017	Asynchronous DNS with a 50 ms Resolution Delay, address sorting and staggered attempts; 250 ms is one recommended default delay.	Pinned reference for expected behaviour. A recommended delay is not a predicted completion time.
RFC 6555 [2], RFC 6724 [3] 2012	Original racing technique [2]; default address-selection policy that decides ordering [3].	Explains why order depends on host policy and reachability.
HE v3 draft -04 [4] 2 July 2026	Adds SVCB/HTTPS DNS record support. Active Internet-Draft, not yet an RFC.	Background only. Pinned revision; no implementation or conformance claims.
Bajpai and Schönwälder [5] ANRW 2016	Historical result (2013–2016 data): a 300 ms timer often selected the slower IPv6 path despite small differences in connect time.	Motivates testing moderately slower IPv6 paths (S3). Not a statement about today's networks.
Sattler et al. [6], [7] IMC 2025	Studied nine browsers on seven operating systems; publishes a testbed framework and a public webtester.	Closest prior work. We add a compact, reproducible AIORI scenario matrix with saved configurations, per-run captures and documented interpretation.

Proposed Solution

One reproducible lab, two clients, a healthy control and six failure or delay scenarios

Lab Linux namespaces; separate IPv4 and IPv6 links plus an unimpaired control path

DNS Controlled test-zone DNS service; AAAA answers can be delayed (S6)

Clients curl and Chromium via Playwright; pinned versions; QUIC and Secure DNS off

Evidence Packet captures plus client and runner logs, joined by run ID

Output Run records, per-scenario statistics and a dashboard

How delays are set: each "RTT" value is a target added round-trip time, split equally across both directions on that family's own link (baseline: 10 ms each way, 20 ms RTT). The resulting RTT is measured before every scenario and recorded separately.

ID	Condition	IPv4	IPv6	What it tests
C0	**Healthy baseline (control)**	baseline	baseline	Normal behaviour
S1	**IPv6 blackhole**	baseline	100% loss	Fallback when IPv6 gets no reply
S2	**IPv6 very slow**	baseline	+600 ms RTT	Fallback under high IPv6 latency
S3	**IPv6 slightly slow**	baseline	+150 ms RTT	Whether slower IPv6 still wins
S4	**IPv4 blackhole**	100% loss	baseline	Behaviour when IPv4 is unavailable
S5	**IPv4 slow**	+600 ms RTT	baseline	Effect of a degraded IPv4 path
S6	**Delayed AAAA answer**	baseline	baseline; AAAA reply +300 ms	Effect of DNS answer timing on connection attempts
X1	**Both degraded (optional)**	+100 ms RTT, 5% loss	+300 ms RTT, 30% loss	Compound stress test, mobile-inspired

Orange rows are mobile-inspired latency conditions: they simulate path latency and are not carrier-network measurements. X1 runs only if time allows.

Optimization Proposed

Consistent metrics and a repeatable design make the two-client comparison interpretable

Design choice	Benefit to this project
Scripted delay and failure profiles	Makes conditions repeatable
At least 20 runs per client and scenario	Reduces dependence on a single observation
Packet capture plus runner logs	Correlates network events with application outcomes
Consistent timing definitions	Makes comparisons interpretable
Scenario-by-scenario analysis	Connects each observation to a labelled, plausible explanation

Illustrative reference for S1 (IPv6 tried first, 20 ms IPv4 RTT): fast-path reference ≈ 20 ms; with a 250 ms attempt delay ≈ 270 ms (≈ 220 ms with curl's documented 200 ms). Reference points only, not predictions or pass/fail thresholds.

Metrics, from packet captures and run logs

1 TCP setup time
First SYN of the run to the ACK that completes the winning handshake. DNS time is recorded separately.

2 Inter-family attempt delay
First SYN of the second family minus first SYN of the first family; “not observed” if only one family is tried.

3 Scenario penalty
A client's median setup time in a scenario minus its own median in C0.

Also reported: winning family, success or failure, failed or abandoned attempts.

Optional sensitivity experiment (before and after)
Before: curl at its documented default (200 ms [8]). After: --happy-eyeballs-timeout-ms varied from 50 to 500 ms on S1–S3. Run only once the core matrix is complete.

Solution Architecture & Design

An orchestrator drives the lab; each run leaves a packet capture and a run log, linked by run ID

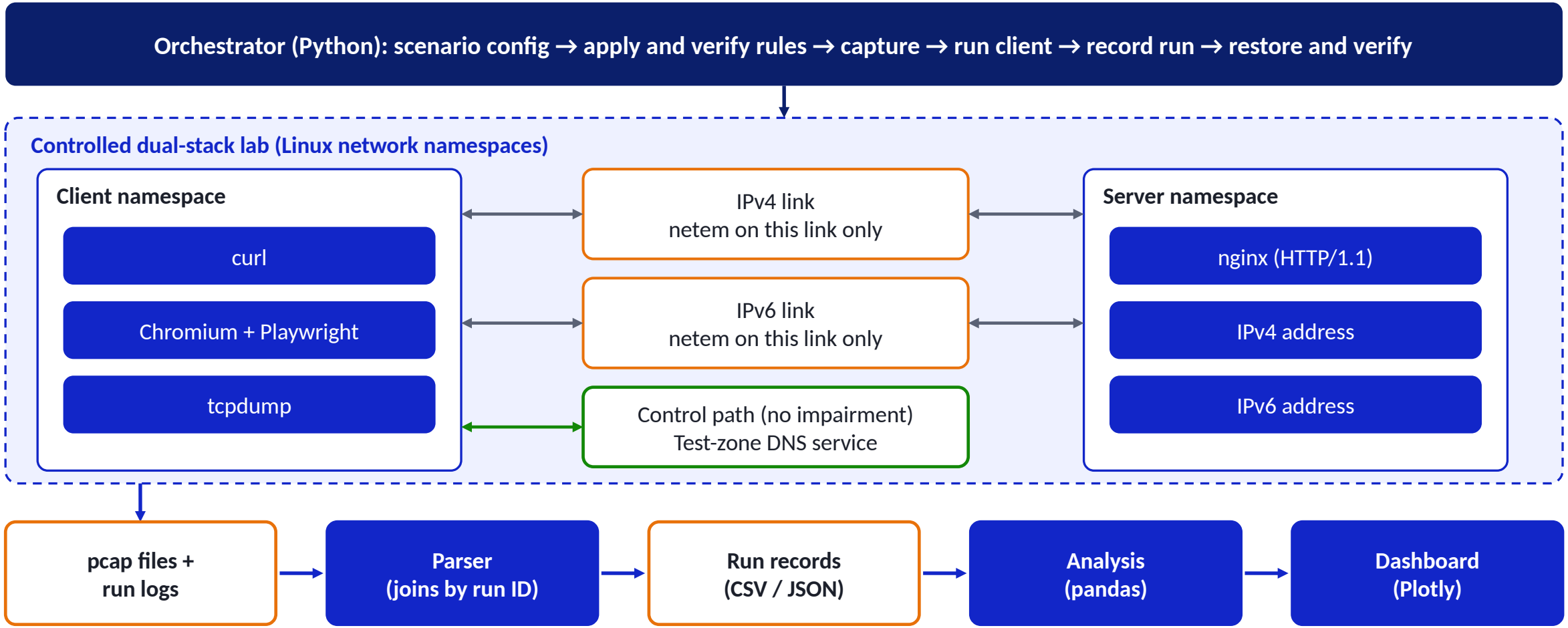
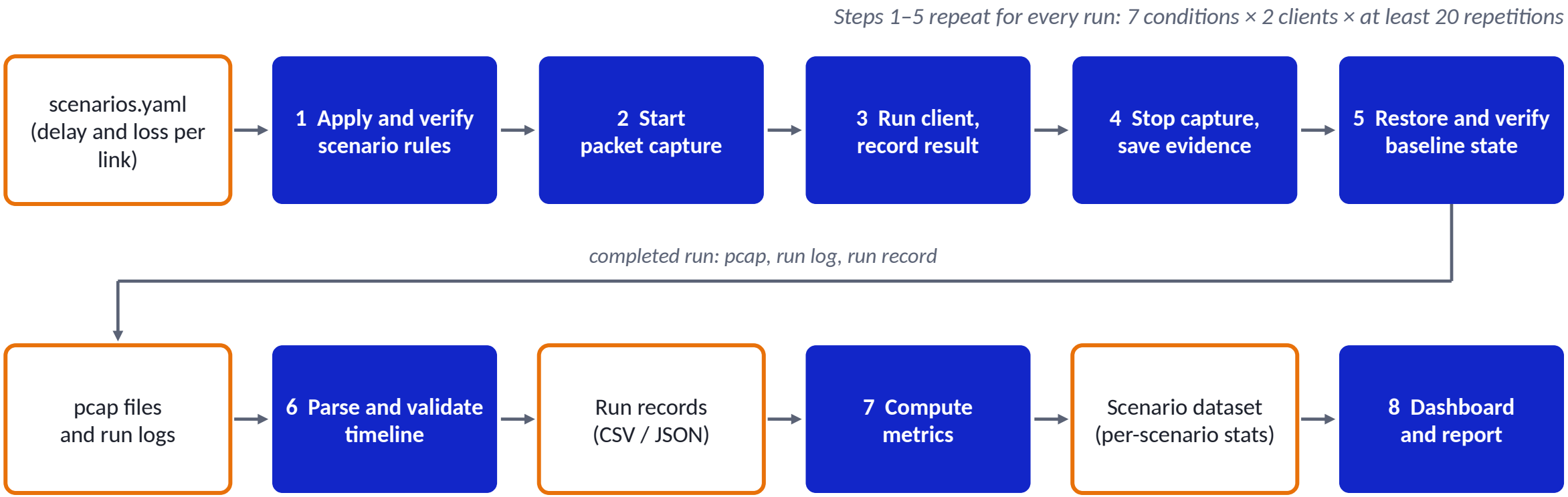


Figure: system architecture. Impairments apply only to the IPv4 and IPv6 links; DNS and orchestration use a separate, unimpaired path.

Data Flow Diagram

Each run follows the same loop and ends by restoring and verifying the baseline



Each run record holds: run ID, scenario, client and version, configuration, timestamps, outcome and packet-capture path. Raw captures are kept unchanged for audit.

Figure: data flow diagram (DFD)

Deployment Timeline

Six dated milestones that follow the AIORI-3 schedule [10]; planned work, no results yet

1. **Oct 11–14 · Level 1 submission** — Sam + Pranav
   Proposal PDF, private repository with ownership transfer, YouTube introduction video
2. **Oct 16–20 · Lab and curl runner** — Sam leads
   Namespaces, dual-stack server, controlled DNS, netem rules, curl runner, first captures
3. **Oct 20–23 · Chromium runner and parser** — Shared; Pranav leads browser
   Playwright runner (QUIC and Secure DNS off), scenario checks, parser checked by hand
4. **Oct 23–26 · Full data collection** — Sam builds, Pranav validates
   C0 and S1–S6 on both clients, at least 20 runs each (280 runs), structured results
5. **Oct 26–29 · Metrics, dashboard, report** — Sam: data; Pranav: UI + QA
   Three metrics, scenario analysis, interactive dashboard and technical report
6. **Oct 30–31 · Final validation and demo** — Sam + Pranav
   Audit files, clean re-run, rehearse demo and pseudocode snippet, complete submissions

Methodology: one factor varies at a time in S1–S6 (the optional X1 is a labelled compound stress test); network rules are reset and verified for every run; client versions, resolver and Secure DNS settings are recorded; packet-level claims are checked against captures, and interpretations state their limits.

Bibliography

Standards, drafts, studies and event documents cited in this proposal

[1] D. Schinazi and T. Pauly, “Happy Eyeballs Version 2: Better Connectivity Using Concurrency,” RFC 8305, IETF, Dec. 2017.

[2] D. Wing and A. Yourtchenko, “Happy Eyeballs: Success with Dual-Stack Hosts,” RFC 6555, IETF, Apr. 2012.

[3] D. Thaler, R. Draves, A. Matsumoto and T. Chown, “Default Address Selection for Internet Protocol Version 6 (IPv6),” RFC 6724, IETF, Sep. 2012.

[4] T. Pauly, D. Schinazi, N. Jaju and K. Ishibashi, “Happy Eyeballs Version 3: Better Connectivity Using Concurrency,” Internet-Draft draft-ietf-happy-happyeyeballs-v3-04 (work in progress), 2 Jul. 2026.

[5] V. Bajpai and J. Schönwälder, “Measuring the Effects of Happy Eyeballs,” in Proc. ACM, IRTF & ISOC Applied Networking Research Workshop (ANRW ’16), Berlin, Jul. 2016, doi:10.1145/2959424.2959429.

[6] P. Sattler, M. Kirstein, L. Wüstrich, J. Zirngibl and G. Carle, “Lazy Eye Inspection: Capturing the State of Happy Eyeballs Implementations,” in Proc. ACM IMC ’25, Madison, WI, Oct. 2025, doi:10.1145/3730567.3732925.

[7] Happy Eyeballs Webtester, source code, github.com/happy-eyeballs/he-webtester (online at happy-eyeballs.net).

[8] curl project, “--happy-eyeballs-timeout-ms,” curl command-line tool documentation, curl.se/docs/manpage.html, accessed 14 Oct. 2026 (documented default: 200 ms).

[9] AIORI-3, “PS-028: Happy Eyeballs IPv4/IPv6 Fallback Measurement,” problem statement, Category: Internet Measurement, 2026.

[10] AIORI-3 Hackathon Guideline, portal.aiori.in/aiori-3/hackathon-guideline/, accessed 14 Oct. 2026.

Thank You for Your Attention!

Questions and live demo

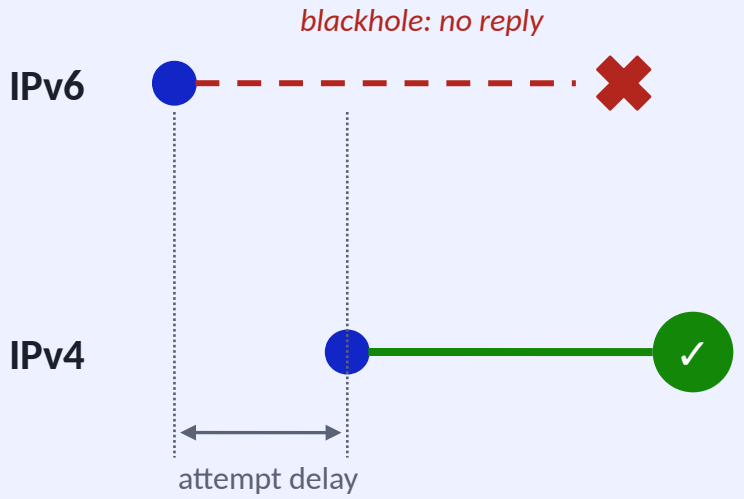
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Who wins the race? (illustration)



The first successful connection is used.
We record what each client actually does.